Stefania Konstantinidi, Ph.D.

Postdoctoral Research Fellow, Department of Mechanical Engineering and
Department of Electrical Engineering and Computer Science
Massachusetts Institute of Technology

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RESEARCH INTERESTS

I build soft actuators where motion is physically encoded into the material itself. By designing fiber-reinforced elastomers, I dictate exactly how the material bends, twists, or contracts. I apply this technology accross mechanically active implants that restore human facial movement, and to bioinspired, morphing robotic wings. Looking ahead, my research vision focuses on next-generation hybrid robotics: embedding fine, distributed soft 'musculature' into conventional robotic systems to actively control, shape, and complement force-generating motors and passively deforming linkages.

EDUCATION

Ph.D., Robotics, Control and Intelligent Systems
École Polytechnique Fédérale de Lausanne (EPFL), Switzerland
Thesis: Reinforced dielectric elastomer actuators for the restoration of facial movement
Advisors: Prof. Yves Perriard, Dr. Thomas Martinez 2021–2025

M.Sc., Robotics
École Polytechnique Fédérale de Lausanne (EPFL), Switzerland
Thesis grade: 6.0/6.0, conducted within C-MedTec, EPFL Innovation Park 2018–2020

B.Sc., Microengineering
École Polytechnique Fédérale de Lausanne (EPFL), Switzerland 2015–2018

ACADEMIC APPOINTMENTS

Postdoctoral Research Fellow, Massachusetts Institute of Technology
Aerial and Underwater Robotics Laboratory (Prof. Raphaël Zufferey) and
Soft and Micro Robotics Laboratory (Prof. Yufeng Chen)
Joint appointment combining aeroelastic flight dynamics with soft micro-actuation; independently funded by the Swiss National Science Foundation. 2025–present

Doctoral Researcher, EPFL
Center for Artificial Muscles and
Integrated Actuators Laboratory 2021–2025

GRANTS AND FELLOWSHIPS

Swiss National Science Foundation, Postdoc.Mobility Fellowship
Fibre reinforced soft and active membranes for deformable flapping wings 2025–present

Sophie Afenduli Foundation Fellowship
Soft actuators for facial reanimation. 2021

Venture Kick and X-Grant - startup funding awards
Diagnostic tool for concussion (C-MedTec). 2019

HONORS AND AWARDS

Invited Rising Star Talk, Northeaster Coloquium of Robotics, Princeton University	2026
Rising Star in Mechanical Engineering, nominated and selected for the workshop in Stanford	2026
Best Poster Award, Hellenic-American Meeting of Early-Career Researchers (HAMER)	2026
Best Demonstrator Award, SPIE EAP/Soft Robotics-in-Action	2026
Best Demonstrator Award (2nd place), IEEE RoboSoft Workshop on Soft Robotic Actuation and Sensing based on Functional Materials	2025
Journal Cover, *Advanced Materials Technologies*	2025
Best Paper Award, IEEE International Conference on Electrical Machines and Systems (ICEMS)	2023
Finalist, Start Lausanne startup competition	2020
iGEM Grand Prize, 353 international teams; additionally Best Presentation, Best Wiki, and Best Human Practices awards	2019

PATENT

1. **S. Konstantinidi**, et al. "Electromechanical device for actuating a patient's eyelid." International Patent Application WO2026028097.

PUBLICATIONS

12 peer-reviewed journal articles (7 as first or co-first author), 1 under review, 7 peer-reviewed conference papers, 22 conference presentations and posters. † *denotes equal contribution.*

Journal Articles

1. **S. Konstantinidi**, N. Liu, R. Ben Mustapha, S. Holzer, M. Koenigsdorff, A. Benouhiba, Y. Civet, Y. Perriard. "Dynamic modeling for the control of reinforced dielectric elastomer actuators for post-paralysis facial movement restoration." *Interface Focus*, 2026.

2. M. Koenigsdorff, S. Holzer, **S. Konstantinidi**, Y. Civet, Y. Perriard. "A soft robotic fish with a dielectric elastomer actuator body and negative stiffness spine." *Advanced Robotics Research*, 2026.

3. M. Koenigsdorff, **S. Konstantinidi**, A. Endesfelder, P. Osipov, J. Mersch, M. Vorrath, M. Zimmermann, Y. Perriard, G. Gerlach. "Elimination of necking and aspect ratio dependence in uniaxial actuators by continuous fiber reinforcement." *Advanced Robotics Research*, 2026.

4. **S. Konstantinidi**, M. Koenigsdorff, A. Osorio Salazar, A. Benouhiba, T. Martinez, Y. Civet, G. Gerlach, Y. Perriard. "Towards artificial muscle implants: structured reinforcement of dielectric elastomers." *Smart Materials and Structures*, 2025.

5. **S. Konstantinidi**, M. Koenigsdorff, P.-J. Martin, A. Benouhiba, Y. Civet, Y. Perriard. "DEyeA: Artificial muscles for the restoration of eye blinking for facial paralysis." *Advanced Materials Technologies*, 2025. [**Cover article**]

6. A. Weinzierl, E. Piccinni, S. Kollarik, **S. Konstantinidi**, Y. Civet, Y. Perriard, P. Giovanoli, N. Lindenblatt. "Current innovations for the treatment of facial nerve paralysis." *JPRAS Open*, 2025.

7. S. Holzer, B. Tiwari, **S. Konstantinidi**, Y. Civet, Y. Perriard. "Design and characterization of an equibiaxial multi-electrode dielectric elastomer actuator." *Materials*, Special Issue on Electroactive Polymers, 2025.

8. A. Benouhiba, S. Holzer, **S. Konstantinidi**, Y. Civet, Y. Perriard. "The elastic frontier: dielectric elastomer actuators in healthcare technology." *Smart Materials and Structures*, 2024.

9. **S. Konstantinidi**†, M. Koenigsdorff†, T. Martinez, A. Benouhiba, J. Mersch, Y. Civet, G. Gerlach, Y. Perriard. "Investigation of buckling instabilities in fiber-reinforced DEAs." *Composites Science and Technology*, 2024.

10. **S. Konstantinidi**†, S. Holzer†, M. Koenigsdorff, T. Martinez, Y. Civet, G. Gerlach, Y. Perriard. "Fiber-reinforced equibiaxial dielectric elastomer actuators for out-of-plane displacement." *Materials*, 2024.

11. **S. Konstantinidi**, B. Tandon, T. Martinez, Y. Civet, Y. Perriard. "Uniaxial reinforced dielectric elastomer actuators with embedded 3D-printed fibers." *Smart Materials and Structures*, 2023.

12. **S. Konstantinidi**, C. Imholz, T. Martinez, A. Benouhiba, A. Walter, Y. Civet, N. Lindenblatt, Y. Perriard. "Real-time actuation of a dielectric elastomer actuator neuroprosthesis for facial paralysis." *Smart Materials in Medicine*, 2023.

In Preparation

1. **S. Konstantinidi**, S. Kollarik, A. Walter, A. Weinzierl, A. Benouhiba, Y. Civet, N. Lindenblatt, Y. Perriard. "In vivo actuation of compliant dielectric elastomer artificial muscles via a peripheral nerve interface." *Preprint.*

2. **S. Konstantinidi**, Y. Chen, R. Zufferey. "Embedding a musculature into flapping wings.".

Peer-Reviewed Conference Papers

1. **S. Konstantinidi**, B. Tankwa, Q. De Menech, T. Martinez, Y. Civet, Y. Perriard. "Post-paralysis upper facial movement restoration in real time using soft actuators." *IEEE Engineering in Medicine and Biology Conference (EMBC)*, 2024.

2. Q. De Menech, L. Andre, **S. Konstantinidi**, A. Benouhiba, T. Martinez, Y. Civet, A. Baldit, Y. Perriard. "Characterisation of polymer materials for the development of an artificial urethra." *IEEE Engineering in Medicine and Biology Conference (EMBC)*, 2024.

3. **S. Konstantinidi**, J. Asboth, A. Walter, S. Holzer, T. Martinez, Y. Civet, Y. Perriard. "An experimental approach for the design of uniaxial fiber-reinforced dielectric elastomer actuators." *IEEE ICEMS*, 2023. [**Best Paper Award**]

4. Q. De Menech, **S. Konstantinidi**, T. Martinez, A. Benouhiba, Y. Civet, Y. Perriard. "Mechanical characterization of porcine urethra: non-linear constitutive models and experimental approach." *IEEE ICABME*, 2023.

5. **S. Konstantinidi**, T. Martinez, A. Benouhiba, Y. Civet, Y. Perriard. "Soft actuators for facial reanimation." *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022.

6. **S. Konstantinidi**, Q. De Menech, A. Walter, P. Mohagheh, T. Martinez, Y. Civet, Y. Perriard. "Characterization of dielectric elastomer actuators: a design of experiments approach." *IEEE ICEMS*, 2022.

7. **S. Konstantinidi**, T. Martinez, R. Mottet, Y. Civet, Y. Perriard. "DEA self-discharge characterization for low duty cycle applications." *IEEE ICEMS*, 2021.

Invited Talks

Université de Cergy Pontoise, France — Hosted by Prof. Cédric Plesse 2026

National Technical University of Athens (NTUA), Greece — Invitation as part of the HAMER award, hosted by Prof. Evangelos Papadopoulos 2026

2nd International Colloquium "Interactive Soft-Stiff Composites: From Basic Research to Practice", Dresden, Germany — *Fiber-reinforced DEAs: exploring anisotropic designs and instabilities* 2024

Annual Meeting of the European Association of Plastic Surgeons (EURAPS), Athens — *Dielectric elastomer actuators for post-paralysis facial reanimation* 2024

Conference Presentations and Posters

1. **S. Konstantinidi**, Y. Chen, R. Zufferey. "Active aeroelastic membranes for bat-inspired flight." *EuroEAP*, 2026.
2. **S. Konstantinidi**, Y. Chen, R. Zufferey. "Bio-inspired membranes for flapping wing robots." *HAMER*, 2026. [**Best Poster Award**]
3. M. Koenigsdorff, S. Holzer, **S. Konstantinidi**, Y. Civet, Y. Perriard. "Combining fiber-reinforced DEAs with a buckling beam system for a soft robotic fish." *SPIE EAPAD*, 2025. [**Best Demonstrator Award**]
4. **S. Konstantinidi**, S. Holzer, A. Osorio Salazar, Y. Civet, Y. Perriard. "Dielectric elastomer actuators for the restoration of facial movements." *IEEE RoboSoft*, 2025. [**2nd Best Demonstrator Award**]
5. **S. Konstantinidi**, S. Holzer, Y. Civet, Y. Perriard. "Dielectric elastomer actuators for the restoration of facial movements." *SPIE EAPAD*, 2025. [Demonstrator]
6. **S. Konstantinidi**, M. Koenigsdorff, S. Holzer, Y. Civet, G. Gerlach, Y. Perriard. "Contractile dielectric elastomer actuators with embedded active and passive structured fibres." *SPIE EAPAD*, 2025.
7. M. Koenigsdorff, C. Böhmer, **S. Konstantinidi**, O. Osipov, J. Mersch, Y. Perriard, G. Gerlach. "Fiber-reinforced strip DEAs with uniform stretch state." *SPIE EAPAD*, 2025.
8. S. Holzer, **S. Konstantinidi**, Y. Civet, Y. Perriard. "Force measurements of planar dielectric elastomer actuators." *SPIE EAPAD*, 2025.
9. **S. Konstantinidi**, S. Kollarik, Y. Civet, A. Weinzierl, Y. Perriard, N. Lindenblatt. "Artificial muscles for facial reanimation." *Swiss Plastic Surgery Congress*, 2024.
10. **S. Konstantinidi**, M. Koenigsdorff, S. Holzer, J. Mersch, Y. Civet, G. Gerlach, Y. Perriard. "Influence of fibre reinforcement on electroactive strain and force in uniaxial DEAs." *ACTUATOR*, 2024.
11. S. Holzer, B. Tiwari, **S. Konstantinidi**, Y. Civet, Y. Perriard. "A dielectric elastomer based miniaturized soft planar microactuator." *ACTUATOR*, 2024.
12. **S. Konstantinidi**, M. Koenigsdorff, A. Benouhiba, J. Mersch, Y. Civet, G. Gerlach, Y. Perriard. "Buckling instabilities in fiber-reinforced DEAs." *EuroEAP*, 2024.
13. M. Koenigsdorff, **S. Konstantinidi**, Y. Civet, Y. Perriard, G. Gerlach. "Investigation of two types of fiber reinforcement for uniaxial DEAs." *EuroEAP*, 2024.
14. **S. Konstantinidi**, P.-J. Martin, A. Benouhiba, Y. Civet, N. Lindenblatt, Y. Perriard. "Artificial muscles for facial reanimation." *EuroEAP*, 2024.
15. **S. Konstantinidi**, Q. De Menech, T. Martinez, P. Germano, A. Boegli, Y. Civet, Y. Perriard. "Inversing the actuation cycle of dielectric elastomer actuators for a facial prosthesis." *SPIE EAPAD*, 2024.

16. **S. Konstantinidi**, M. Koenigsdorff, T. Martinez, Y. Civet, G. Gerlach, Y. Perriard. "Frequency response of fiber-reinforced DEAs." *SPIE EAPAD*, 2024.

17. M. Koenigsdorff, J. Mersch, **S. Konstantinidi**, Y. Perriard, G. Gerlach. "Highly anisotropic carbon fiber electrodes for DEAs and their dynamic non-monotonic conductive properties." *SPIE EAPAD*, 2024.

18. S. Holzer, A. Walter, **S. Konstantinidi**, T. Martinez, Y. Civet, Y. Perriard. "Carbon-based printed electrodes for DEAs: a study of pad, inkjet and stencil printing." *SPIE EAPAD*, 2024.

19. Q. De Menech, S. Zammouri, **S. Konstantinidi**, A. Benouhiba, Y. Civet, Y. Perriard. "An artificial urinary sphincter based on dielectric elastomer actuators." *SPIE EAPAD*, 2024.

20. **S. Konstantinidi**, C. Imholz, T. Martinez, A. Benouhiba, A. Walter, Y. Civet, Y. Perriard, N. Lindenblatt. "Dielectric elastomer actuators for facial paralysis." *Swiss Plastic Surgery Congress*, 2023.

21. **S. Konstantinidi**, A. Thabuis, T. Martinez, Y. Civet, Y. Perriard. "Contractile dielectric elastomer actuators with embedded structured fibers." *EuroEAP*, 2023.

22. **S. Konstantinidi**, T. Martinez, Y. Civet, Y. Perriard. "Fiber-reinforced dielectric elastomer actuators for in-plane contraction displacement." *EuroEAP*, 2022.

RESEARCH COLLABORATIONS

Eight cross-institutional collaborations spanning clinical surgery, materials science, and photonics; initiated and coordinated by me unless noted otherwise.

Therapeutic Technology Device & Development Lab at MIT — Prof. E. Roche. 2026–ongoing
Dielectric elastomer actuators for implantable and assistive medical devices.

Music Technology Lab at MIT — Prof. M. Rau. Soft actuator-based laryngeal implant for the restoration of vocal-fold mobility. 2026–ongoing

Technische Universität Dresden, Institute of Solid-State Electronics — Prof. G. Gerlach. Joint investigation of instabilities in fiber-reinforced dielectric elastomer actuators; produced four co-authored publications and an invited talk. 2023–2025

University Hospital Zurich (USZ), Department of Plastic Surgery — Prof. N. Lindenblatt. Clinical partnership covering prosthesis design, surgical constraints, and the design and execution of in vivo animal trials. 2021–2025

ETH Zurich — Combining soft actuation with topology optimization to design structured reinforcement architectures. 2024

EPFL Microsystems Laboratory (LMIS1) — Advanced additive manufacturing for fiber reinforcement of soft membranes: inkjet printing and melt electrowriting. 2023

Swiss Center for Electronics and Microtechnology (CSEM), Neuchâtel — Exchange of additive manufacturing methods and ink materials. 2023

Biomedical Optics Research Laboratory, University Hospital Zurich — Prof. M. Wolf, and **Advanced Quantum Architecture Laboratory, EPFL** — Prof. E. Charbon. Cross-institutional collaboration bridging clinical near-infrared spectroscopy and photonic detector engineering for a non-invasive concussion diagnostic; supported finite element modeling of mechanical impact and the associated biological cascade, and validation against an anatomically precise head phantom (C-MedTec). 2019–2020

RESEARCH MENTORING AND SUPERVISION

Supervised **20 research projects** by 20 students across EPFL, USZ and MIT. * indicates a project that resulted in a peer-reviewed publication with the student as a co-author.

Master's Theses

Enrico Bregni*	MIT	2026	Design of a pterosaur-inspired flapping wing
Pierre-Jean Martin*	EPFL	2024	Design of an eye prosthesis
Carlotta Imholz*	USZ	2022	Real-time actuation of DEAs

Research Interns

Raki Ben Mustapha*	EPFL	2024	Closed loop control of DEAs
Baptiste Tankwa*	EPFL	2023	Post-paralysis upper facial movement restoration

Semester Projects

Cali Hendricks	MIT	ongoing	Onboard high voltage supply for aerial robots
Mariia Maliar	MIT	ongoing	Bat-inspired flapping mechanism
Jad Benabdelkader	EPFL	2025	Machine learning modeling and control of DEAs
Myriam Charfeddine	EPFL	2024	Design of a bistable element
Leen Bou Khaled	EPFL	2024	Structured auxetic fiber-reinforced actuators
Andy Bittan-Mosseri	EPFL	2024	Dielectric elastomer actuators for facial reanimation
Pierre Lorthiois	EPFL	2024	Inkjet printing of fiber-reinforced actuators
Lucas Romano	EPFL	2024	Inkjet printing of fiber-reinforced actuators
Mischa Oleh Mez	EPFL	2024	Design of a humanoid facial robot
Chiara Evangelisti	EPFL	2024	Design and modeling of a tissue surrogate
Arnaud Sansonnens	EPFL	2023	Modeling of fiber-reinforced DEAs
Lucas Muespach	EPFL	2023	Design and development of a humanoid facial robot
Sarah Soussou	EPFL	2023	Inkjet-printed dielectric elastomer actuators
Julian Asboth*	EPFL	2023	Uniaxial fiber-reinforced DEA fabrication
May Salama	EPFL	2022	Characterization of DEAs' self-discharge

TEACHING

Teaching Assistant, EPFL — *Programming (C/C++)*, *Electronics*, *Actuators and Electromagnetic Systems*. Led exercise sessions and laboratory instruction; designed and graded assessments. 2021–2025

EPFL Tutoring Attestation — formal certification in university-level teaching 2023

Prepared to teach: mechatronics and actuator design; soft robotics; smart materials and electroactive polymers; finite element modeling; introductory programming and electronics.

PROFESSIONAL SERVICE AND LEADERSHIP

Treasurer, MIT Postdoctoral Association — administer an annual budget exceeding $35,000; negotiated a 25% budget increase with the MIT Office of the Vice President 2026–present

Session Chair, Materials Research Society (MRS) Fall Meeting, Boston — Symposium SF05, *From Robotic Toward Autonomous Materials*; Symposium EN02, *Next-Generation Technologies for Clean Energy and Materials Sustainability* 2025

Reporter, Materials Research Society (MRS) Fall Meeting — selected to cover symposium sessions and report on emerging research for the society's meeting coverage 2025–present

Outreach and Dissemination Committee, EuroEAP Society (European Scientific Network for Soft Transducers and Electromechanically Active Polymers) 2024–present

Journal Reviewer — *Science Advances*; *Advanced Materials*; *IEEE Robotics and Automation Letters*; *Smart Materials and Structures*; *Materials Research Express* 2023–present

Operational Firefighter, SDIS Etraz, Switzerland — trained in hydraulics and radio communications 2021–2025

Event Manager, Treasurer, and Partnership Manager, Erasmus Student Network, EPFL — administered an annual budget exceeding CHF 120,000; organized weekly programming for up to 200 students 2016–2020

OUTREACH AND MEDIA

Swissnex in Boston and New York, Invited panelist on "How robotics are reinventing medicine" 2026

MIT Museum, Presentation and exhibition of working prototypes to visiting public 2026

MExpo, MIT Mechanical Engineering Exposition — exhibited bioinspired and soft robotic systems to the MIT community and visiting public 2026

Advanced Science News — *Artificial muscle could help facial paralysis sufferers blink again*, coverage of the *DEyeA* study 2025

Werner Siemens-Stiftung — feature article on the facial reanimation implant, *An artificial muscle for a real smile* 2024

EPFL News — *EPFL Bioengineering Team Wins Global iGEM Competition* 2019

EPFL Open Days; regular presentation of research to visiting students and industry partners 2021–2025

ADDITIONAL PROFESSIONAL EXPERIENCE

Investment Fellow, Verve Ventures, Switzerland
Evaluated medical technology ventures at a European early-stage investment platform. 2023

Research and Development Leader, C-MedTec, EPFL Innovation Park
Launched a medical startup for concussion diagnosis using near-infrared spectroscopy. Developed finite element models of mechanical impact and the associated biological cascade; validated results using an anatomically precise head phantom. Selected by the VP for Innovation and VP for Education to conduct a master's thesis within the venture. 2019–2020

Research and Development Engineer, iGEM, Switzerland and USA
Developed *ViTest*, a field-deployable diagnostic for the grapevine diseases flavescence dorée and bois noir, delivering results in under two hours with no laboratory equipment or specialist training. Responsible for device design and microneedle fabrication for plant DNA extraction; contributed to the team wiki, market research, and human practices. 2018–2019

TECHNICAL SKILLS AND LANGUAGES

Experimental: soft actuator fabrication (DEAs, elastomer processing, electrode deposition by pad/inkjet/stencil printing), fiber-reinforced composite fabrication, mechanical and electromechanical characterization, in vivo animal trial design and execution, microfabrication, conventional machining.

Computational: finite element modeling, nonlinear constitutive modeling of soft materials, real-time

control implementation, machine learning for actuator modeling, C/C++, Python, MATLAB, CAD.

Languages: Greek and French (native/bilingual); English (C2 Proficiency); German (B1); Spanish and Russian (A2); Mandarin Chinese (HSK2).

INTERESTS

Endurance sports and outdoor pursuits, including marathon and trail running (top 100 women in Switzerland for marathon times), skiing: alpine/backcountry (training for the 2028 Patrouille des Glaciers), climbing, sailing (MIT Ladies' Team), windsurfing, and fishing (2nd place, international competition, Montreux 2024)

REFERENCES

Prof. Raphaël Zufferey
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Prof. Nicole Lindenblatt
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Prof. Gerald Gerlach
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